

Jai Pannu

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Education

University of Waterloo | Honours Mechatronics Engineering, Co-op Waterloo, ON | Sep. 2025 – Present

Technical Skills

Technical Skills: C++, Python, BACnet, STM32, CAN, FreeRTOS, PID Control, HMI Development, SolidWorks, AutoCAD, Excel/VBA

Experience

Controls and Automation Engineering Intern | Voyager Controls Mississauga, ON | May. 2026 – Present

- Configured 130+ BACnet controllers for closed-loop FCU control across commercial BAS deployments.
- Designed control points for 30+ AHUs, RTUs, dampers, and other HVAC systems, applying heat transfer and fluid flow fundamentals.
- Developed operator HMI screens in ABB doDESIGN for BAS control and remote automation.
- Performed low-voltage functional testing and troubleshooting of building control systems and field devices.
- Authored over 6 thermostat installation SOPs aligned with Ontario HVAC safety requirements; built VBA templates that reduced control-drawing time by 60%.

Testing & Evaluation Engineer | AfterQuery Experts (YC W25) San Francisco, CA | May. 2026 – Present

- Evaluated and calibrated dToF LiDAR, IMU, and stereo-depth sensors for humanoid and UGV perception.
- Documented 10+ end-effector configurations in reinforcement-learning environments and produced technical analyses on physical intelligence.

Extracurriculars

Firmware Developer | University of Waterloo Formula Electric Waterloo, ON | Jan. 2026 – Present

- Developed STM32 firmware for a digital potentiometer using SPI and UART for control-system integration on the Formula SAE electric race car.
- Integrated FreeRTOS+CLI with reusable peripheral structures, reducing firmware scaling effort by 30% across the Formula SAE vehicle platform.

Humanoid Firmware Designer | WATonomous Humanoid Autonomy Waterloo, ON | Sep. 2025 – Dec. 2025

- Integrated a CubeMars AK80-9 elbow actuator into a 51.2 V humanoid power system and shared 7-actuator CAN bus via CANable.
- Established actuator command/feedback and safe operation using CAN termination and E-stop protection, enabling 22 N·m peak torque.

Projects

Commissioned Autonomous UGV Rover, Tracer Golf

- Sponsored by Tracer Golf to design and commission an autonomous golf-ball retrieval rover and create a replication build kit for non-technical staff.
- Built a reproducible <\$900 COTS UGV using LiDAR SLAM, odometry, IMU sensor fusion, and pose-graph mapping.
- Integrated ESKF, AMCL reinitialization, fiducials, and map-consistency recovery, achieving <5 cm / 1–2° repeatable localization across 100+ runs.

AlbertaSat CubeSat Electrical Power System

- Soldered and integrated wiring for deployable solar wings on flight-bound CubeSat hardware.
- Built fixtures and test rigs for repeatable assembly, connection inspection, continuity testing, and functional verification.

Sub-150 g Competition Fighting Robot

- Integrated a 3100 KV BLDC weapon motor, dual 600 RPM drive motors, ESCs, receiver, and 2S LiPo within a strict 150 g mass limit.
- Redesigned the motor mount to a 12.2 mm M2 interface and simulated the BLDC in Ansys Electronics before manufacturing.

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Mechatronics Engineering at the University of Waterloo



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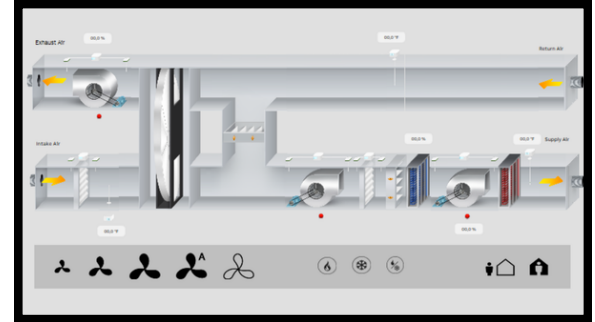
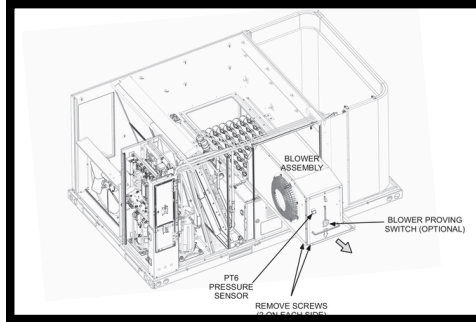
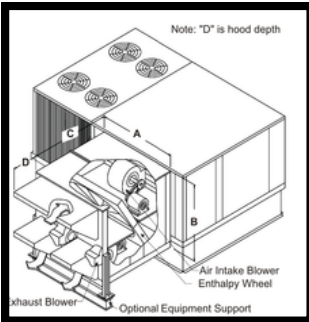
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Voyager Controls Co-op

Voyager Controls is a leading Building Automation Systems contractor in the HVAC industry. Voyager designs, assembles, programs, and installs field controllers and PLCs, and structured cabling, for industrial and construction settings.



What?

- On-site team lacked clear instructions on controller installation
- Control shop drawings required multiple redundant steps
- Clients lack an intuitive way to control their installed Building Automation Systems

How?

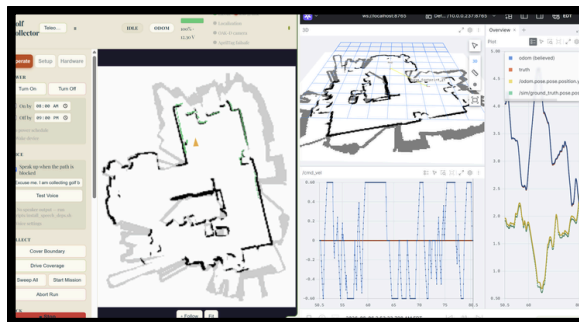
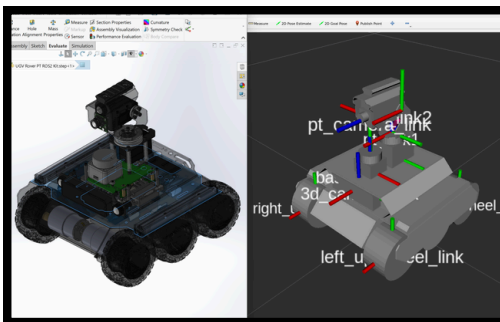
- Created over 6 detailed video tutorials on configuring controllers with safety priority
- Created **VBA**-driven component templates for easy drawing completion and tighter schematic checking
- Designed control points for **30+** HVAC components and programmed their panel interface with **ABB doDesign**

Results

- Clear documentation with consistent site installations
- Reduced the time taken from drawing start to finish by **60%**
- Improved client satisfaction with a friendly interface mapped to the actual component points for remote control and automation

Commissioned Autonomous UGV Golf-Picking Rover

Tracer Golf is Canada's first fully automated indoor driving range and golf simulator facility. As a robotics enthusiast I had the pleasure of being sponsored design a rover to make the process of retrieving balls autonomous and creating the build kit/documentation required for non-technical staff to replicate the units for multiple facilities. (Limited by NDA)



What?

- Indoor golf facilities allow a private experience for customers to enjoy golf with minimal intervention as most tasks are automated and allow self-service
- Facilities are mostly operated by non-technical staff that prioritize quick delivery and automation instead of products that require much preparation

How?

- Designed a UGV rover built on the WaveShare platform that can be reproduced with COTS parts that are **<\$900**
- Used **LIDAR SLAM**, odometry, and **IMU** sensor fusion to allow mapping via teleoperation and stored using pose-graph SLAM
- Integrated error-state Kalman filter, AMCL, particle filter reinitialization, fiducial based-localization, and map consistency checks as localization recovery mechanisms

Results

- Built a complete 2D facility map with repeatable **<5 cm/1-2°** localization accuracy across **100+ runs**
- Deployed a web interface linked to secure WLAN for operator access
- Delivered documentation/build kit for replication of autonomous rovers across Tracer facilities

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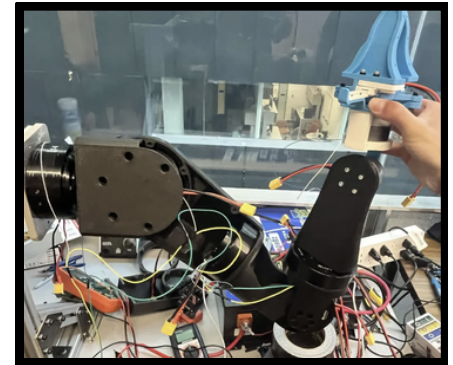
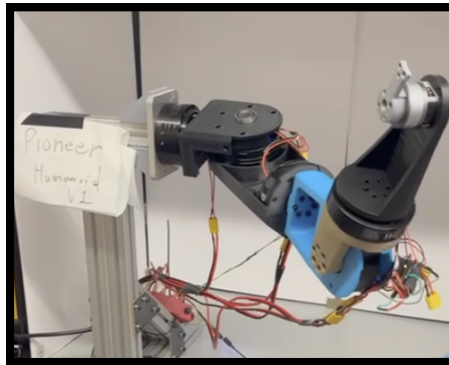
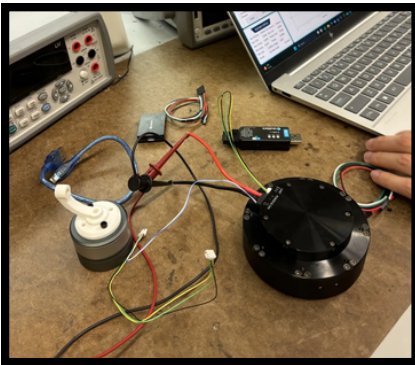
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Humanoid Elbow Actuation

WATonomous is the University of Waterloo's flagship autonomous robotics design team. I worked on the Humanoid robot sub-team where I specialized in firmware and vision components.

WATO



What?

- Humanoid elbow required reliable power, communication, and control
- **CubeMars** actuator needed integration with the arm's shared electrical system
- Motor commands and feedback had to connect with the main software stack

How?

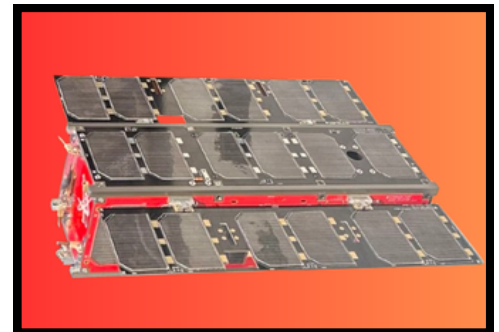
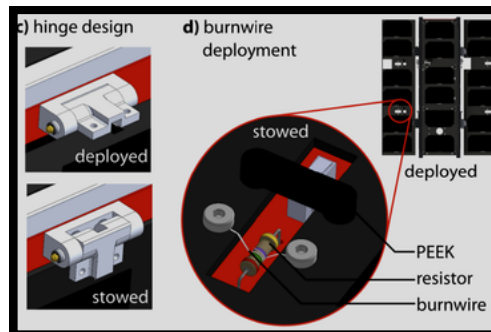
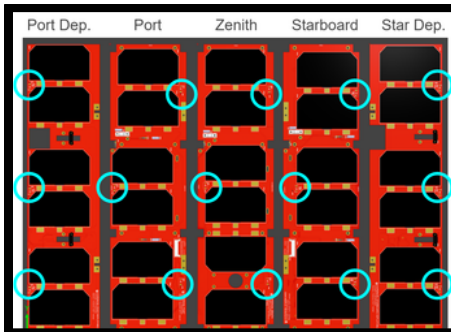
- Integrated the elbow's CubeMars **AK80-9** actuator into the 51.2 V power architecture
- Connected the shared **CAN** bus via a **CANable adapter** for actuator command and feedback
- Supported safe operation using **E-stop protection** and **CAN termination**

Results

- Delivered 22 N·m peak torque for reliable elbow actuation
- Achieved **44.9 N·m/kg** torque density in a compact 490 g actuator
- Integrated the elbow into the arm's shared 7-actuator CAN network

Albertasat Electrical Power System

AlbertaSat develops Cube Satellites under the CSA's CUBICS program, being at the frontier of Canada's student-led space mission programs. AlbertaSat was the first Albertan organization to launch fully student-made functional satellites, followed by multiple successful missions. I joined their work as a high-schooler and contributed as a firmware team member in manufacturing and installing the Hyperion solar wings. ([Link](#))



What?

- CubeSat solar wings required reliable electrical integration with the spacecraft
- Flight hardware required precise assembly and repeatable verification

How?

- **Soldered** and integrated wiring for the CubeSat's deployable solar wings
- Built and used **fixtures** and **test rigs** to support repeatable assembly
- Inspected connections and assisted with electrical continuity and functional testing

Results

- Completed solar-wing electrical assembly for the flight-bound CubeSat
- Contributed hands-on **electrical manufacturing** and validation experience to an orbital spacecraft

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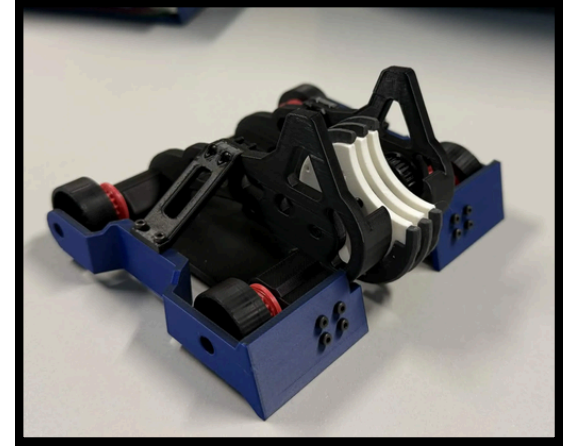
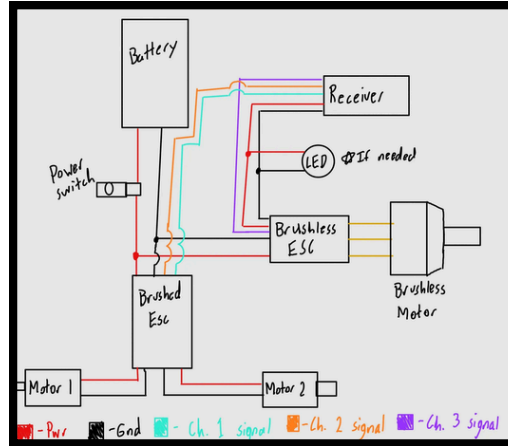
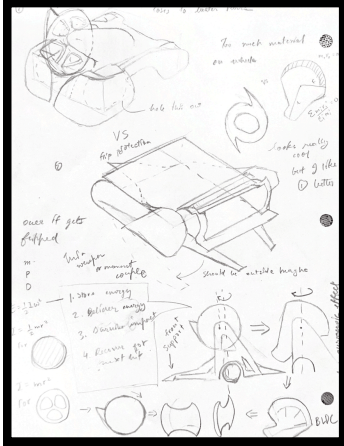


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Sub 150g Competition Fighting Robot



What?

- Designed a competitive robot within a strict 150 g weight limit
- Needed compact drive, weapon, power, and control systems in one chassis
- Weapon assembly required adaptation to fit the available brushless motor

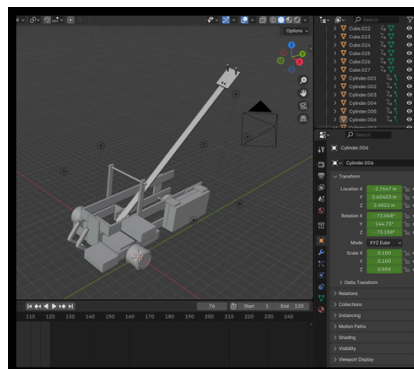
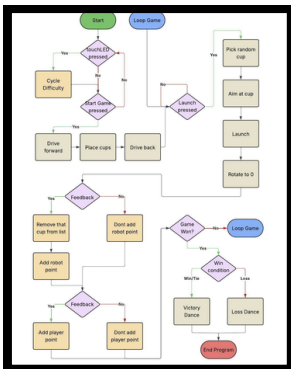
How?

- Modified the CAD for a **3100 KV brushless DC** weapon motor
- Redesigned the motor mount to a 12.2 mm bolt circle with M2 fasteners
- Integrated dual **600 RPM** drive motors, **ESCs**, receiver, switch, and 2S LiPo
- Used **Ansys Electronics** to simulate the BLDC before manufacturing

Results

- Resolved motor incompatibility using a 12.2 mm M2 mounting interface
- Completed end-to-end electromechanical integration, from CAD to physical hardware

KOBE - Beer Pong Playing Robot (link)



What?

- An autonomous VEX-based robot capable of playing beer pong by launching projectiles into targets at fixed distances
- Applied the **engineering design lifecycle** and streamlined development process by leveraging **technical drawings**.

How?

- Engineered a closed-loop **PID** controller in **C++** to autonomously drive the robot and maintain a precise firing distance
- Programmed location verification using encoder feedback
- Simulated pre-manufactured product in **Blender** for designing

Results

- Achieved an **86% shooting accuracy** (6 out of 7 hits) during final performance trials
- Executed full intake-to-fire cycles in an average of 13.6 seconds, successfully meeting design constraints for speed and reliability